From Gaussian Samples to a Complex Wishart Minimum Law in Lean

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Abstract

We present a Lean formalization of selected complex Gaussian Gram matrix laws, with a complete actual-model chain for the smallest eigenvalue of a central complex 2×2 Wishart matrix. The construction starts from independent real standard Gaussian coordinates, fixes the unit complex variance convention, and derives an independent Bartlett coordinate law. An explicit four-real-coordinate chart yields the normalized Gram density on the positive definite cone. Translation of this cone, followed by a Hermitian spectral threshold criterion, identifies the actual minimum law as exponential with rate two. We also formalize its positive scalar variance dependence and a weakest-mode logarithmic-rate outage formula. Supporting developments include noncentral one-column results, an arbitrary-row one-column small-threshold asymptotic, and fixed-subspace Gaussian projection and independence interfaces. We distinguish these actual probability laws from general-dimensional candidate-kernel identities and unresolved joint spectral densities. The mathematical distribution formulas are classical; the contribution is their explicit formal connection to an underlying Gaussian sample model and the reusable interfaces needed for that connection. Independent semantic review and release replication remain pending in this working version.

1 Introduction

Wishart eigenvalues occur in multivariate probability and in the analysis of multiple-antenna channels. In eigenmode transmission, an eigenvalue determines the gain of a scalar subchannel. Jin, McKay, Gao, and Collings [1] give formulas for ordered eigenvalue distributions of complex noncentral Wishart matrices and apply them to communication performance. This paper studies the formal verification of selected probability statements underlying that analysis.

The main difficulty is connecting representations. A Gaussian sample can be represented by real coordinates, complex columns, a Gram matrix, coordinates on the Hermitian space, and ordered eigenvalues. A density identity at one level does not automatically establish the law at another. Variance factors, the reference measure, exceptional sets, and strict versus weak events must be tracked through each transition. We organize the development around equalities of pushforward measures so that a proposed density cannot enter as an assumed description of the model it is meant to identify.

Our principal example is a central complex matrix $H \in \mathbb{C}^{2\times 2}$ with independent unit-variance entries. We prove the coordinate density of H^*H and derive

$$\mathbb{P}\{\lambda_{\min}(H^*H) > x\} = e^{-2x}, \qquad x \geq 0.$$

The proof uses a translation of the positive definite cone and does not require an ordered spectral Jacobian. Positive scalar covariance and a logarithmic-rate statistic are then obtained by transporting events through this actual law.

The contribution has three parts. First, the central 2×2 chain exposes the distributional, coordinate, and spectral obligations separately. Second, the project provides reusable Gaussian projection and Hermitian-coordinate interfaces, together with a dimension-independent noncentral one-column asymptotic. Third, it supplies a declaration-level account of what has been checked and what remains outside the formalized scope. We make no claim that the classical distribution formulas are new, or that this is the first formalization of Wishart theory. A comprehensive novelty comparison remains part of release preparation.

The development builds on Lean and mathlib [2]. In particular, the change-of-variables infrastructure developed in mathlib [3] and its Gaussian and independence libraries carry much of the foundational analysis. The role of probability kernels in this ecosystem is described by Degenne [4]. Our contribution specializes and connects those interfaces to concrete Gaussian matrix statistics; it does not replace their underlying general theorems.

2 Model, measures, and formal scope

Let U_{ij}, V_{ij} be independent $N(0,1)$ variables and set

$$H_{ij} = \frac{U_{ij} + \mathrm{i}V_{ij}}{\sqrt{2}}, \qquad X = M + H, \qquad W = X^*X. \tag{1}$$

Thus $\mathbb{E}|H_{ij}|^2 = 1$. The complex scalar density is $\pi^{-1}\exp(-|z|^2)$ relative to $d\Re z\, d\Im z$. The sample space in Lean is a finite-dimensional real Euclidean space named `ComplexSample`; `complexSampleMatrix` performs the complex encoding. The underlying law is mathlib's `stdGaussian`. The deterministic mean M is added after encoding. The Gram law is a measurable pushforward, rather than a measure defined initially by a Wishart density formula.

For a general $t \times s$ sample we use X^*X when $s \le t$. For a one-column sample, the only Gram eigenvalue is $\|X\|_F^2$. For the main example, $s = t = 2$ and $M = 0$. Eigenvalues are ordered decreasingly; the last coordinate is the minimum. The formal real-valued statistic `twoRowActualMinimum` is computed from the actual sample Gram matrix.

2.1 The Hermitian reference measure

Write

$$W = \begin{pmatrix} u & b + \mathrm{i}c \\ b - \mathrm{i}c & v \end{pmatrix}. \tag{2}$$

The reference measure is $du\, db\, dc\, dv$. The positive definite coordinate cone is

$$D = \{(u, b, c, v) : u > 0,\ b^2 + c^2 < uv\}.$$

It is a four-dimensional real measure on Hermitian matrices. Restricting eight-dimensional Lebesgue measure on all complex 2×2 matrices to the Hermitian subspace would produce a different, unusable reference measure.

In dimension s, `CentralHermitianCoordinates` constructs an explicit measurable equivalence between s^2 real coordinates and the Hermitian subspace. Its coordinate measure is a product of real Lebesgue measures, transported by this equivalence. This construction fixes a normalization; it does not by itself identify any Gaussian Gram density.

2.2 Meaning of the verification claims

An actual-law theorem starts with the sample measure in (1) and identifies its pushforward. A coordinate theorem gives an equivalence, a Jacobian, or a transformation of integrals. A candidate identity simplifies a proposed analytical expression. These are recorded separately. In particular, a theorem about a candidate integral does not count as a probability theorem until its density has been derived from the sample.

The named project declarations discussed below have kernel-check evidence. The repository's printed axiom checks use only `propext`, `Classical.choice`, and `Quot.sound` for the audited declarations. This is a dependency statement, not a proof that the mathematical interpretation of every definition matches an external paper. Independent review of model conventions and analytical specifications remains pending.

Human semantic review required. The Gaussian encoding, the Hermitian coordinate measure, and the identification with the communication model require independent scrutiny.

3 Actual Bartlett coordinates and the Gram density

Let h_1, h_2 be the columns of the central 2×2 sample. Set $S = \|h_1\|^2$. On the event $h_1 \neq 0$, normalize the first column, project the second column onto it, and write the resulting complex coefficient as $Z = A + \mathrm{i}B$. Let E be the squared norm of the orthogonal residual. The implementation gives a measurable unit-direction fallback at zero. It subsequently proves that the exceptional first-column event has mass zero.

Theorem 3.1 (Actual Bartlett law)**.** *The sample-defined coordinates satisfy*

$$\mathcal{L}(S, Z, E) = \mathrm{Gamma}(2, 1) \otimes \mathcal{CN}(0, 1) \otimes \mathrm{Exp}(1),$$

where Gamma and exponential parameters are rates. Relative to $ds\,da\,db\,de$, their density is

$$f_B(s, a, b, e) = \frac{s}{\pi} e^{-(s+a^2+b^2+e)}\, \mathbf{1}_{\{s>0, e>0\}}. \tag{3}$$

Proof. For each fixed encoded first column, the projection and residual calculation gives the same $\mathcal{CN}(0, 1) \otimes \mathrm{Exp}(1)$ fiber law. The direction map is measurable and defined everywhere. The established Gaussian skew-product law transports this fixed-fiber statement to the actual two-column sample, showing independence from the first column. Its energy has the already established $\mathrm{Gamma}(2, 1)$ law. Mapping this joint law to (S, Z, E) gives the displayed product measure. The densities se^{-s}, $\pi^{-1}e^{-(a^2+b^2)}$, and e^{-e} multiply to (3). The positive-domain restriction is derived from these actual marginal laws, including the absence of boundary mass. □

This step is stronger than establishing independence for a fixed projection. The first-column direction is random, so measurability and the invariant fiber law must be used explicitly. The corresponding principal declarations are `actualTwoRow_BartlettCoordinates_law` and `actualTwoRow_BartlettRealCoordinates_eq_normalizedDensity`.

The Gram coordinate chart is

$$T(s, a, b, e) = (s, \sqrt{s}\, a, \sqrt{s}\, b, a^2 + b^2 + e). \tag{4}$$

Its domain is $s > 0, e > 0$, and its image is D. The inverse on D is

$$(u, b, c, v) \longmapsto \left(u, \frac{b}{\sqrt{u}}, \frac{c}{\sqrt{u}}, v - \frac{b^2 + c^2}{u}\right).$$

The formal chart is proved to reproduce the actual Bartlett Gram mapping. It is not an unrelated parameterization with an assumed connection to H.

Proposition 3.2 (Jacobian and actual Gram density)**.** *On the positive Bartlett domain,* $|\det DT| = s$*. The coordinate law of* H^*H *has density*

$$f_W(u,b,c,v) = \pi^{-1}e^{-(u+v)}\mathbf{1}_D(u,b,c,v). \tag{5}$$

Its integral is one.

Proof. The derivative of (4), in the stated coordinate order, is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ a/(2\sqrt{s}) & \sqrt{s} & 0 & 0 \\ b/(2\sqrt{s}) & 0 & \sqrt{s} & 0 \\ 0 & 2a & 2b & 1 \end{pmatrix},$$

whose determinant is s. Lean verifies the Fréchet derivative, the determinant, the inverse, and the image of the domain. The change-of-variables theorem therefore divides the Bartlett density by s. The factor s in (3) cancels, and the exponent becomes $-(u+v)$. The actual-coordinate pushforward and its almost-sure positive support identify the resulting measure with the actual Gram law. Normalization follows from that law being a probability measure; it is not an additional density premise. □

The main declaration is `actualTwoRow_GramRealCoordinates_eq_traceDensity`; the existing Wishart-measure API is exposed by `complexWishart_twoRow_realCoordinates_eq_traceDensity`.

Human semantic review required. Product-measure transport, complex-to-real volume normalization, the orientation of the Gram coordinates, and the inverse Jacobian are retained as explicit items in the independent review packet.

4 From the positive cone to the minimum law

For a Hermitian matrix W,

$$\lambda_{\min}(W) > x \quad \Longleftrightarrow \quad W - xI \succ 0. \tag{6}$$

The project proves this spectral criterion and its concrete four-coordinate version. It uses the smallest coordinate of the decreasing ordered spectrum.

Theorem 4.1 (Central complex 2×2 minimum)**.** *For the actual sample* (1) *with* $M = 0$ *and* $s = t = 2$*, and* $x \geq 0$*,*

$$\mathbb{P}\{\lambda_{\min}(H^*H) > x\} = e^{-2x}, \tag{7}$$

$$\mathbb{P}\{\lambda_{\min}(H^*H) \leq x\} = 1 - e^{-2x}. \tag{8}$$

For $x < 0$ *the CDF is zero. The complete minimum law is* $\mathrm{Exp}(2)$*.*

Proof. By (6) and Proposition 3.2, the tail is the integral of $\pi^{-1}e^{-\operatorname{tr} W}$ over $W \succ xI$. For $x \geq 0$ this region lies in the positive definite support. Substitute $W = V + xI$. Translation preserves the four-real-coordinate measure, while $\operatorname{tr}(V + xI) = \operatorname{tr} V + 2x$. Hence the integral is

$$e^{-2x}\int_{V\succ 0}\pi^{-1}e^{-\operatorname{tr} V}\,dV = e^{-2x},$$

using the normalization already derived from the actual Gram law. The complement of the strict tail event is exactly the weak CDF event; no exchange of strict and weak inequalities is needed here. Positive semidefiniteness of a Gram matrix gives the negative-threshold case. Finally, uniqueness of probability measures with a common CDF identifies the whole pushforward measure with Exp(2). □

The three principal statements are `actualTwoRow_minimum_tail`, `actualTwoRow_minimum_CDF`, and `actualTwoRow_minimum_law_expTwo`. The unmodified determinant candidate of Jin et al. at $(s, t, L) = (2, 2, 0)$ is also shown equal to this actual CDF in `theorem1CentralTwoRow_candidate_eq_actualCDF`. This last connection is a specialization of their formula, not a proof of their general ordered eigenvalue distribution theorem.

4.1 Positive scalar covariance

Corollary 4.2 (Scale convention)**.** *Let $a > 0$, $v = a^2$, and $W_a = (aH)^*(aH)$. Then*

$$\lambda_{\min}(W_a) = a^2 \lambda_{\min}(H^*H), \qquad \mathcal{L}(\lambda_{\min}(W_a)) = \mathrm{Exp}(2/a^2).$$

In particular, the CDF for variance $v > 0$ is $1 - e^{-2x/v}$ for $x \geq 0$.

Proof. The actual encoded sample matrix respects real scalar multiplication. Consequently its Gram matrix scales by a^2. The spectral threshold criterion identifies the corresponding minimum scaling, and transporting the events in Theorem 4.1 gives the CDF. CDF uniqueness again gives the full law. □

Here a is an amplitude and v is an entry variance. The result addresses a positive scalar covariance; it does not address arbitrary correlated columns.

5 An information-theoretic event interface

Consider the mathematical statistic

$$C_{\min}(H; a, \rho) = \log(1 + \rho \lambda_{\min}((aH)^*(aH))), \qquad a > 0, \quad \rho > 0,$$

with natural logarithms. Under a scalar Gaussian eigenmode model this is the usual per-mode logarithmic-rate expression, measured in nats; the communication background is described by Tse and Viswanath [5]. The formal theorem concerns this explicitly defined statistic. It does not formalize a coding theorem, optimal power allocation, or the total MIMO mutual information $\log\det(I + \rho H^*H)$.

Corollary 5.1 (Weakest-mode log-rate outage)**.** *For $R \geq 0$,*

$$\mathbb{P}\{C_{\min}(H; a, \rho) \leq R\} = 1 - \exp\left(-\frac{2(e^R - 1)}{\rho a^2}\right). \tag{9}$$

For $R < 0$, the probability is zero.

Proof. For every $g \geq 0$ and $\rho > 0$, positivity of $1 + \rho g$ and monotonicity of the exponential give

$$\log(1 + \rho g) \leq R \quad \Longleftrightarrow \quad g \leq \frac{e^R - 1}{\rho}.$$

For $R \geq 0$ the transformed threshold is nonnegative, so the actual scaled CDF applies. For $R < 0$, $C_{\min} \geq 0$ pointwise makes the event empty. □

The actual probability statement is `actualTwoRow_weakestLogRate_outage`. Its classical interpretation is deliberately separated from the proved event transformation. A rate specified in bits uses $R \log 2$ in place of R; the present Lean statement uses nats.

Human semantic review required. The operational interpretation of ρ and the allocation of power among modes must be specified before identifying this statistic with a particular physical outage experiment.

6 Reusable probability foundations and supporting results

6.1 Fixed orthogonal Gaussian decompositions

Let G have the standard Gaussian law on a finite-dimensional real Hilbert space E, and let K be a fixed linear subspace. Write P_K and $P_{K^\perp}$ for the subspace-valued orthogonal projections.

Proposition 6.1 (Actual orthogonal split)**.** *The projected variables have their subspace standard Gaussian laws and satisfy*

$$\mathcal{L}(P_K G, P_{K^\perp} G) = \gamma_K \otimes \gamma_{K^\perp}.$$

If $\dim_{\mathbb{R}} K = 2k$ *with* $k > 0$*, then* $\|P_K G\|^2/2 \sim \mathrm{Gamma}(k, 1)$*.*

Proof. The characteristic function of a linear image is obtained by composing with the adjoint. For a coisometry the adjoint preserves norm, so the standard Gaussian characteristic function is unchanged. This identifies the projected laws. Joint Gaussianity follows by linear transport, and the cross covariance vanishes by orthogonality. The existing Gaussian independence theorem then gives independence and the product law. The norm statement transports the established Euclidean radial Gamma law through an orthonormal coordinate isometry; it does not require a new radial integration proof. □

These interfaces are `stdGaussian_orthogonalProjectionOnto_law`, `stdGaussian_orthogonal_split_law`, and `stdGaussian_projected_half_norm_sq_law`. They reuse mathlib's Gaussian characterization and independence machinery. The word *fixed* is essential: substituting a random earlier-column subspace requires a separate measurable conditional-law argument.

6.2 Noncentral one-column small-threshold behavior

Theorem 6.2 (Arbitrary positive row count)**.** *For an integer* $m > 0$*, a deterministic* $M \in \mathbb{C}^{m\times 1}$*, and the sample convention* (1)*, let* $F_M(x) = \mathbb{P}\{\|M + H\|_F^2 \le x\}$*. Then*

$$\lim_{x \downarrow 0} \frac{F_M(x)}{x^m} = \frac{e^{-\|M\|_F^2}}{m!}. \tag{10}$$

Proof. Let μ be the real encoded mean, so $\|\mu\|^2 = 2\|M\|_F^2$. The actual event is a real Gaussian ball of radius $r = \sqrt{2x}$. For its centered density φ, translation gives

$$\varphi(z - \mu) = e^{-\|\mu\|^2/2 + \langle z, \mu\rangle} \varphi(z).$$

Cauchy–Schwarz bounds the inner product on that ball by $\pm r\|\mu\|$. Integrating yields

$$e^{-\|M\|_F^2 - r\|\mu\|} F_0(x) \le F_M(x) \le e^{-\|M\|_F^2 + r\|\mu\|} F_0(x).$$

The central energy has the already proved $\mathrm{Gamma}(m, 1)$ law, with $F_0(x)/x^m \to 1/m!$. Dividing the inequalities by x^m and applying the squeeze theorem proves (10). □

The theorem is `noncentralOneColumnSmallestCDF_div_tendsto_frobenius`. It includes zero mean and does not assume a noncentral spectral density. Exact actual-model formula connections for one-column row counts one, two, and three are also present, with the noncentrality parameter matched to the Frobenius mean norm. They are supporting coverage, not a general multi-column noncentral result.

7 Verification, diagnostics, and reproducibility

7.1 Declaration-level evidence

The pinned toolchain is Lean `v4.35.0-rc2`; the mathlib revision is `e18afff334217588afc719a8b707b00d90521064`. The repository's `lean-toolchain` and `lake-manifest.json` fix these dependencies. No toolchain upgrade is part of the reported development. Table 1 gives module entry points; the companion claims manifest records exact declarations and source hashes.

Result	Principal source module(s)
Actual coordinate density	`TwoRowActualDensity`; `TwoRowActualNormalizedDensity`
Spectral tail and full CDF	`TwoRowMinimumLaw`
Original central candidate	`Theorem1CentralTwoRowActualCDF`
Positive scalar covariance	`TwoRowScaledMinimum`
Log-rate outage	`TwoRowLogRateOutage`
Projection and independent split	`GaussianOrthogonalProjectionLaw`; `GaussianOrthogonalSplitLaw`
Noncentral one-column asymptotic	`T4NoncentralOneColumnAnyRows`
General coordinate cone integral	`CentralHermitianCoordinates`; `CentralWishartConeIntegral`

Table 1: Entry points for the reported statements. The last row is a candidate-kernel identity, not an actual general Wishart law.

Three verification levels are kept distinct: elaborating a leaf module, elaborating the aggregate project entry point, and rebuilding all project sources into fresh output. A cached entry-point check is not a clean rebuild. The publication tooling records fresh-source re-elaboration separately from a full Lake build and does not claim to rebuild mathlib from source. All 164 project modules, counting generated submodules and the aggregate entry, passed this fresh-output source check. One stale theorem identifier in an unreferenced four-row support file was repaired without changing its statement; the interrupted run was then resumed using only unchanged successful outputs. This is one completed source-check campaign, not two independent reproductions. The executable status is in the accompanying verification report; a release must retain that report together with its exact source hashes.

The source audit removes nested comments and string literals before searching for prohibited proof placeholders or new axiom declarations. Lexical absence alone is insufficient: declaration-level axiom dependencies and reproducible kernel checking provide the stronger evidence. Independent semantic review is an additional obligation, particularly for the correspondence between formal objects and the intended probabilistic model.

7.2 Numerical controls

Numerical calculations are used to reject specification mistakes before formal proof work, and as controls for the computational interpretations. They are not proof substitutes. The new log-rate

preflight uses seed 20261015 and 200000 actual independent complex Gaussian 2×2 samples. It checks the trace against Gamma$(4, 1)$, the minimum against Exp(2), and the Hermitian quadratic eigenvalue formula against a numerical eigensolver. The corresponding Kolmogorov–Smirnov statistics are approximately 0.00238 and 0.00124; the maximum eigenvalue-formula discrepancy is below $5 \cdot 10^{-15}$.

Fifteen combinations of amplitude, per-mode coefficient, and target rate verify the event transformation and compare its empirical probability with (9). The reported tolerance is eight binomial standard errors plus $8/N$, not a formal simultaneous confidence guarantee. The script and its JSON output preserve the seed, model, controls, and individual results.

7.3 A bounded normalization diagnostic

A separate audit examines the project's transcription of a partition formula associated with Theorem 3 of [1]. A Lean witness proves that the current transcribed central candidate has real part greater than one at $(s, t, L, k, x) = (2, 3, 0, 2, 2)$, so it cannot be an event probability at those arguments. This is an exact statement about a specified expression in the repository.

Independent rank-one experiments at $(s, t, L) = (3, 4, 1)$ use seed 20261015 and 200000 actual samples for each of two noncentrality values. At noncentrality one and threshold one, the empirical partition probability is 0.009510; the displayed-expression implementation gives 0.019397228, while division by the residual factor $\Gamma_2(3) = 2$ gives 0.009698614. Bessel and hypergeometric implementations agree, and trace-law controls pass. Rare-event thresholds have limited Monte Carlo resolution.

These observations do not establish an erratum to the published paper. The journal version, possible conventions, and the exact intended ordered domain still require source reconciliation. We preserve the original candidate and distinguish displayed and derived normalizations. No general Theorem 3 probability bridge is claimed. This diagnostic is ancillary to the main minimum-law contribution.

7.4 AI assistance and independent review

The development used Codex for proof decomposition, implementation, numerical controls, and drafting. The local Ollama model `deepseek-prover-v2:7b` was used as a bounded proof-completion tool for short, precisely stated lemmas. Accepted generated proof bodies retain their provenance and the label `GUIDED_COLLABORATION` where applicable. This workflow is not a benchmark: attempts were not sampled under a controlled comparison protocol, and no general model-performance conclusion is drawn.

Local prover responses did not supply an adequate independent adversarial review. A review packet identifies measure changes, Fubini uses, parameter matching, and strict/weak spectral events for review by a different model family and a mathematically qualified human. No such completed review is claimed in this version. AI tools are not listed as authors; the final human authors must verify and take responsibility for the mathematical and bibliographic content, consistently with arXiv's tool-disclosure policy [6].

8 Generalization boundary and research direction

For $0 < s \le t$, the desired actual central Gram density relative to a fixed s^2-real-coordinate Hermitian volume is

$$K_{s,t} \det(W)^{t-s} e^{-\operatorname{tr} W} \mathbf{1}_{W \succ 0}, \qquad K_{s,t} = \left(\pi^{s(s-1)/2} \prod_{j=1}^{s} (t-j)! \right)^{-1}. \tag{11}$$

The project has *not* derived (11) from the actual Gaussian sample in arbitrary dimension. What is formalized is the translation identity for the corresponding candidate-kernel integral. If the actual density and normalization are subsequently established, the translation would give, for $x \ge 0$,

$$\mathbb{P}\{\lambda_{\min} > x\} = e^{-sx} K_{s,t} \int_{V \succ 0} \det(V + xI)^{t-s} e^{-\operatorname{tr} V} \, dV.$$

The determinant factor must be retained when $t > s$. The kernel identity itself can be stated for negative x; its interpretation as a tail probability then requires the original positive-support indicator and cannot simply be reused unchanged.

The next substantive step is a general Cholesky density induction, rather than a collection of further low-dimensional charts. Four obligations are explicit. First, measurable parameterized projections must lift the fixed-subspace laws to the random earlier-column conditional setting. Second, for $r > 0$ and $z \in \mathbb{C}^{s-1}$ the block map

$$(r, z, B) \longmapsto \begin{pmatrix} r & \sqrt{r}\, z^* \\ \sqrt{r}\, z & B + zz^* \end{pmatrix}$$

must have a dimension-independent real-coordinate absolute Jacobian r^{s-1}. Small-dimensional symbolic controls are available but are not a general Lean proof. Third, the determinant and trace identities must combine with this Jacobian and the recurrence

$$K_{s,t} = \frac{K_{s-1,t-1}}{\pi^{s-1}(t-1)!}.$$

Finally, exceptional sets and full-rank support must be connected to the actual pushforward measure. These obligations identify reusable probability and multivariate-calculus contributions beyond the present example.

General noncentral ordered spectral densities and the full parameter range of the original Theorems 1–4 remain outside the established results. In particular, the present cone method does not supply a noncentral unitary integration formula. The current work supplies a verified example and interfaces for pursuing those questions, rather than a completed formalization of the whole source paper.

9 Conclusion

The development connects real Gaussian samples to an actual complex Gram density and then to a smallest-eigenvalue law, with scalar covariance and logarithmic-rate event transport made explicit. Its main reusable lesson is the separation of actual pushforward laws, coordinate transformations, and candidate formulas. The arbitrary-row one-column asymptotic and fixed orthogonal Gaussian split provide further foundations for a general density induction. Release preparation must still complete independent semantic review, the related-work assessment, author metadata, and a public immutable artifact reference.

A Artifact organization and release conditions

The companion repository places Lean sources under `theory/jin_wishart_formalization/`, numerical controls under `experiments/`, and exact claims and verification reports under `verification/`. This manuscript is self-contained LaTeX; its bibliography is included below, with no external images, custom class, or local input files. The files `claims.json`, `REPRODUCIBILITY.md`, and `SUBMISSION_READINESS.md` accompany the working manuscript.

Before submission, the author line must be completed, an accessible repository or archival artifact URL must be supplied, fresh-source evidence must be checked against the release commit, and the independent review items must be resolved or explicitly narrowed. A final related-work search must support the precise contribution claims. The T3 diagnostic may be omitted if source reconciliation is not complete. arXiv moderation is not peer review [6]; formatting compliance does not imply acceptance.

References

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